

A-4 - Installation Procedures

This section describes what to do when the FastBack® Model 90E arrives at your facility. The following topics are addressed in this section:

- Installing the Drive Assembly
- Installing the Pan Assembly
- Final Positioning
- Wiring the Photoeye
- Calibrating the Photoeye
- Wiring the Power Module
- Installation Checklist

Installing the Drive Assembly

The drive assembly is shipped in a crate that consists of five plywood panels and a base pallet.

Tools Needed: Phillips bit, drill, socket set, wrench, forklift

<u>Step</u>	<u>Description</u>
--------------------	---------------------------

1	Remove top of crate
----------	----------------------------

Using the drill, loosen the screws from the top of the shipping crate and remove the top (see Photo 4-1).



Photo 4-1 – Removing Screws at Top of Shipping Crate

A-4 – INSTALLATION PROCEDURES

<u>Step</u>	<u>Description</u>
-------------	--------------------

1	Remove top of crate (continued)
----------	--

Instructions for disassembling the crate may be written inside the crate, as shown in Photo 4-2.



Photo 4-2 – Instructions Inside Crate

2	Remove ends and sides of crate
----------	---------------------------------------

Using the drill, loosen the screws holding the ends of the crate and remove the ends (see Photo 4-3).



Photo 4-3 – Removing Ends of Crate

Loosen the screws holding the sides of the crate, and remove the sides.

A-4 – INSTALLATION PROCEDURES

FastBack® Model 90E Technical Manual

Step	Description
-------------	--------------------

3	Unbolt articulating feet
----------	---------------------------------

Unbolt the articulating feet from the pallet making up the bottom of the crate (see Photo 4-4).

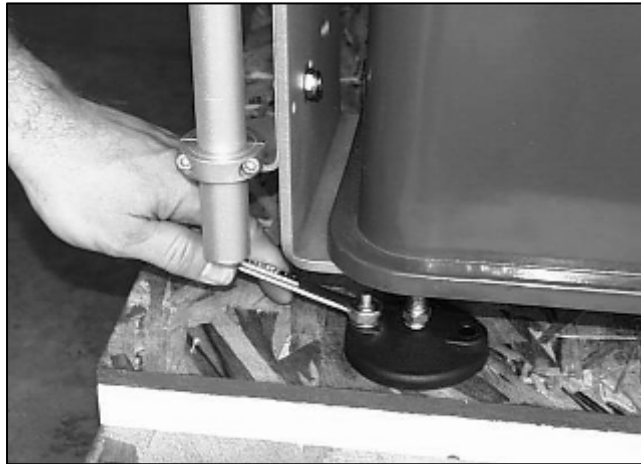


Photo 4-4 – Unbolting the Articulating Feet

4	Prepare to lift drive assembly off pallet
----------	--

Position the forklift between the articulating feet and the baffles on the underside of the drive assembly (see Photo 4-5).



Photo 4-5 – Positioning the Forklift



Make sure that the forks clear the power module cord and the motor junction box cord grip (see Photo 4-6).

A-4 – INSTALLATION PROCEDURES

<u>Step</u>	<u>Description</u>
4	Prepare to lift drive assembly off pallet (continued)



Photo 4-6 – Fork Clearing Cord and Cord Grip

Note: *If your FastBack® arrived with the pan attached to the drive assembly, there is no need to remove the pan. If the pan was shipped separately, do not attach it until the drive assembly is positioned.*

The drive assembly can also be lifted using a forklift and straps, as shown in Photo 4-7.



Photo 4-7 – Lifting Drive Assembly with Straps



If using this lifting method, remove the pan first to avoid damaging the pan.

<u>Step</u>	<u>Description</u>
-------------	--------------------

5	Move drive assembly into position
----------	--

Carefully lift the drive assembly off the pallet and place it in its designated location.

Note: *The drive assembly must be properly oriented front to rear. Refer to the equipment layout drawing for the proper orientation.*

6	Attach drive assembly to support member
----------	--

Loosely bolt the drive assembly to the appropriate support member (T-pedestal or angle iron).

7	Verify shipment is complete (if original shipment)
----------	---

If you are uncrating the original shipment, the drive assembly crate also contains a custom timing plate (part no. 30547820) (see Photo 4-8). The timing plate is used when replacing the eccentric timing belt.



Photo 4-8 – Timing Plate inside Shipping Crate

Note: *Only one timing plate is included with the original shipment. Heat and Control recommends storing the timing plate with your facility's other tools or spare parts.*

A-4 – INSTALLATION PROCEDURES

<u>Step</u>	<u>Description</u>
-------------	--------------------

7	Verify shipment is complete (if original shipment) (continued)
---	---

The original shipment also contains accessories for communicating with, diagnosing, and reprogramming the power module. These accessories are as follows:

Semipower

- One communications cable and adapter with 9-pin serial port connector
- One 9-pin to 25-pin serial port adapter
- One 3½ -inch disk containing the wInControl™ program
- One 3½-inch disk containing configuration files for the inverter
- One DriveBlok owner's manual

Yaskawa

- One hand-held operator with attached communications cable
- One Yaskawa owner's manual

Note: *Only one of each accessory is included with the original shipment. Heat and Control recommends storing these accessories with your facility's other tools or spare parts.*

Photo 4-9 shows the power module accessories as they may have been packed in the crate.



Photo 4-9 – Power Module Accessories in Crate

If any of these items are missing, contact your project manager or the Heat and Control Galesburg, IL office at (309) 343-5518.

Installing the Pan Assembly

Do not install the pan until the drive assembly has been moved into position as described in “Installing the Drive Assembly.”

Note: *If the drive assembly was shipped with the pan installed, proceed to “Final Positioning” later in this section.*

<u>Step</u>	<u>Description</u>
--------------------	---------------------------

1	Lift pan from crate
----------	----------------------------

Remove the lid of the pan assembly shipping crate. Holding the pan in the middle, carefully lift the pan from the crate.

2	Place pan on pan attachment disks
----------	--

Place the pan onto the pan attachment disks on top of the drive assembly.

Place one hand in the bottom of the pan between the two pairs of band clamps and push downward. This will prevent the pan from tipping down.

3	Latch band clamps
----------	--------------------------

Beginning with the rear band clamps, latch each band clamp to fasten the pan to the pan supports (see Photo 4-10). Make sure that the band clamp completely encircles both the top and the bottom pan attachment disks.



The pan may be damaged if it tips before the clamps are latched. Latching the rear clamps first will prevent the pan from tipping.



Photo 4-10 – Supporting Pan While Latching

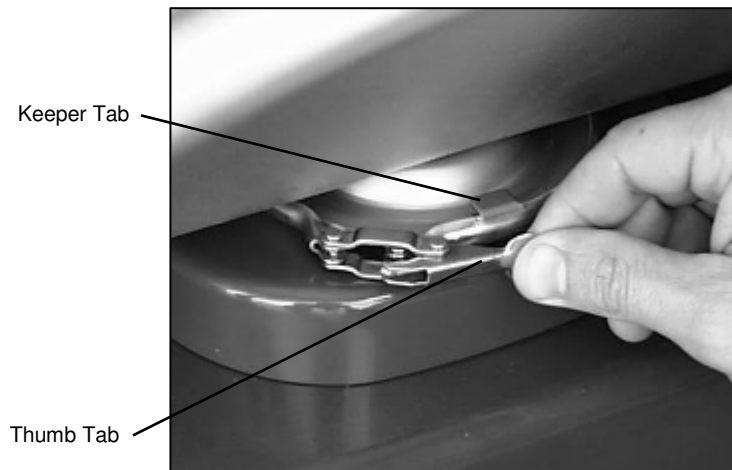
A-4 – INSTALLATION PROCEDURES

FastBack® Model 90E Technical Manual

<u>Step</u>	<u>Description</u>
-------------	--------------------

3	Latch band clamps (continued)
----------	--------------------------------------

Do not let the thumb tab interfere with the slider cover. The keeper tab of the band clamp should cover the notch in the pan attachment disk, making the latched thumb tab parallel to the slider cover. Photo 4-11 shows a correctly latched band clamp.



*Photo 4-11 – Correctly Latched Band Clamp
(Thumb Tab Parallel to Slider Cover)*



If the keeper tab does not cover the notch in the pan attachment disk, the slider cover will rub the top of the enclosure, adding stress to the drive mechanism. Photo 4-12 shows a band clamp that is incorrectly latched.

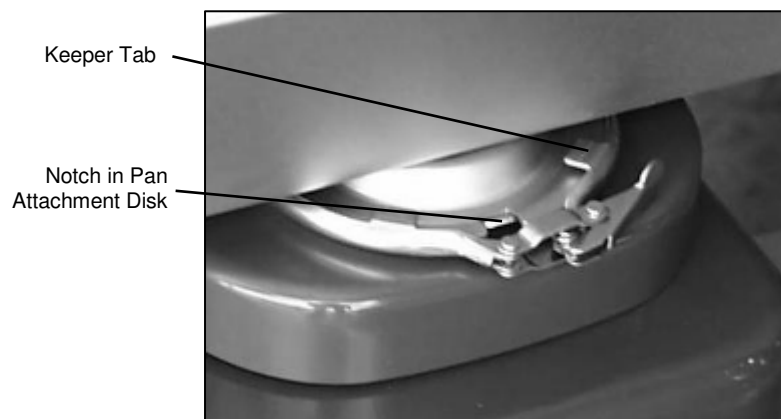


Photo 4-12 – Incorrectly Latched Band Clamp

<u>Step</u>	<u>Description</u>
-------------	--------------------

4	Reinstall adjustable nose discharge if necessary
----------	---

The adjustable nose discharge, if included, may have been reversed for shipping (see Photo 4-13).



Photo 4-13 – Shipping Position of Adjustable Nose Discharge

Reinstall the adjustable nose discharge as follows:

1. Fully loosen the four-prong knobs on each side of the pan.
2. Slide the adjustable nose off the pan.
3. Turn the nose around and slide it back onto the pan. Make sure that the nose is between the two white washers on the four-prong knob (see Photo 4-14)



Photo 4-14 – Correctly Installed Adjustable Nose Discharge

4. Retighten the four-prong knobs.

A-4 – INSTALLATION PROCEDURES

FastBack® Model 90E Technical Manual

Final Positioning

This procedure describes how to perform a final check of the FastBack® Model 90E's position.

Note: *The equipment layout drawing for your facility's specific installation supersedes the general guidelines below.*

Tools Needed: Level, 3/4-inch crows foot wrench, torque wrench, 9/16-inch and 3/4-inch wrench

<u>Step</u>	<u>Description</u>
--------------------	---------------------------

1	Check center alignment
----------	-------------------------------

Verify that the center line of the drive and pan assemblies lines up with the center of the scale in plan view.

2	Check alignment front to rear
----------	--------------------------------------

Verify that the rear of the pan is positioned so that it catches the product from the infeeding equipment. The end of the pan, not counting the adjustable nose, should be 4 inches from the dispersion cone on the scale, from side view (see Figure 4-1).

Note: *The adjustable nose provides ± 2 inches of front-to-rear adjustment.*

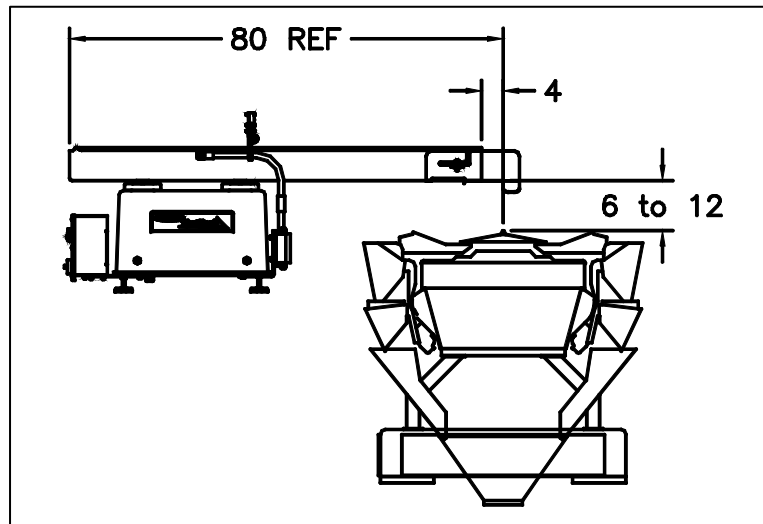


Figure 4-1 - Correct Front-to-Rear Alignment

Step	Description
------	-------------

3	Check horizontal alignment
---	-----------------------------------

Align the pan in side view, according to the dimensions on the equipment layout drawing. The nominal horizontal alignment is 0°. If your facility is a washdown environment, the pan can be declined 0.5° to allow drainage (see Figure 4-2). Adjust the articulating foot using the 9/16-inch wrench.

Note: *Inclining the pan more than 1.5° may impede travel rates. Declining the pan more than 5° will exceed the angle of articulation of the articulating feet.*

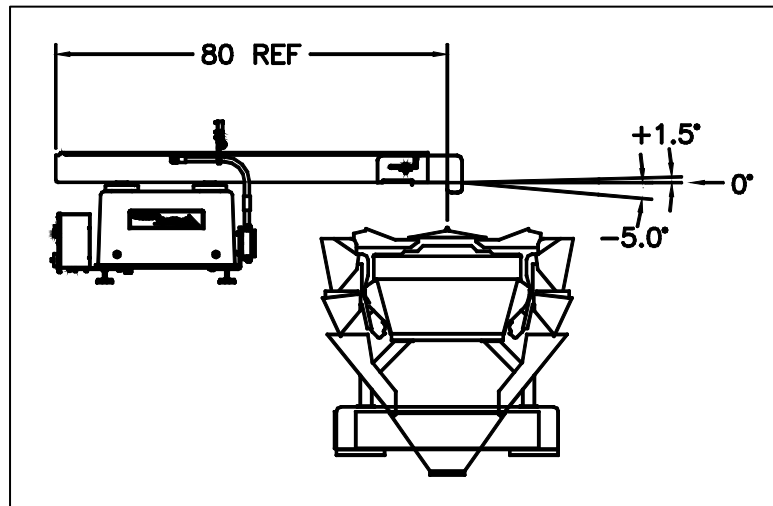


Figure 4-2 - Correct Horizontal Alignment

4	Check vertical alignment
---	---------------------------------

Verify that the bottom of the pan is 6 to 12 inches, according to the equipment layout drawing, from the tip of the dispersion cone on the scale (see Figure 4-1, above). Adjust the articulating feet if necessary, using the 9/16-inch wrench.

Note: *The articulating feet extend 2 inches vertically, ±1 inch from the bottom of the base plate, with ± 5° of swivel.*

5	Tighten articulating feet
---	----------------------------------

After verifying that the FastBack® is properly positioned, use the ¾-inch crows foot wrench and torque wrench to tighten the jam nuts on the articulating feet to 42 foot-pounds. Use the 9/16-inch wrench to keep the threaded stem from turning.

A-4 – INSTALLATION PROCEDURES

Wiring the Power Module

This procedure describes how to connect the FastBack® power module to the primary power and control sources at your facility.

Both the Semipower and Yaskawa power modules come from the factory programmed to run at 50 Hz in single-speed, start/stop mode with a run confirmation signal.

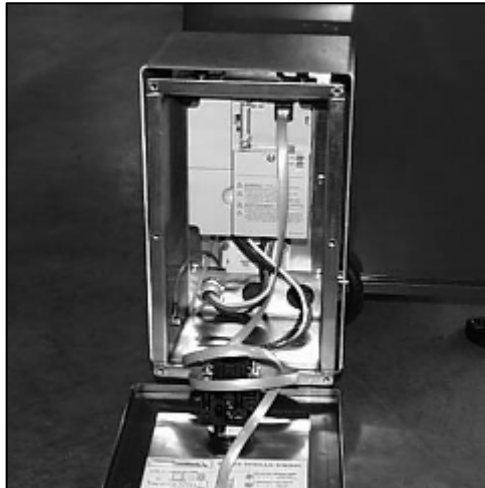
<u>Step</u>	<u>Description</u>
--------------------	---------------------------

1	Open power module box
----------	------------------------------

Remove the six screws holding the lid of the power module box in place.

2	Install cord grips
----------	---------------------------

The back of the power module box contains two open knockouts for wiring primary power and control leads. A third knockout contains the motor leads in a liquid-tight (NEMA 4) cord grip with a seal nut, which is installed at the factory (see Photo 4-21).



*Photo 4-21 – Power Module Box with
Motor Leads Installed*

Install 3/4-inch NTP cord grips in the two open knockouts. For washdown environments, use liquid-tight (NEMA 4) cord grips with seal nuts, along with seal-flex conduit or a multi-conductor, single-jacket wire through each cord grip.

Run the control leads through the center knockout. Run the primary power leads through the remaining knockout. Provide a 2-foot service loop from the primary power cord grip to the emergency disconnect switch, so that the power module lid can be moved out of the way when performing future maintenance.

Step Description

3 Connect primary power leads

Connect primary power as follows:

- Connect the primary power leads to terminals 1, 3, and 5 of the red and yellow emergency disconnect switch.
- Connect the incoming ground wire to the grounding lug on the inside wall of the power module box.

Technical Publication PH009 contains wiring diagrams for the Yaskawa inverter and Technical Publication PH010 contains wiring diagrams for the A-B PowerFlex 70 inverter (mounted in control panel).

4 Connect control leads

Connect the control leads as described in Table 4-2 (for Yaskawa power modules). Wiring diagrams for the Yaskawa V1000 inverter appear in TechPub PH009.

Application	Terminal/Connection
Start/stop All applications (2 wires)	Install a dry contact closure (relay or switch) between terminals SC (digital common) and S1 (run forward).
Single speed Speed select (no wires needed)	No additional wiring needed; conveyor will run at speed set in parameter D1-01 (50 Hz).
Multi-speed Preset speed select (2 wires)	Install a pair of wires to provide dry contact closure (relay or switch) between terminals SC and S5 (preset speed reference 1) and/or SC and S6 (preset speed reference 2). Refer to the preset speed table in the wiring diagram within TechPub PH009 to determine the applicable binary combination.
Variable speed 4-20mA analog input (shielded/twisted pair)	Connect leads (must be a shielded/twisted pair) to AC (analog common) and A2 (frequency reference). A hardware and parameter change is needed to complete the installation. See TechPub PH009 for details.
Run confirmation All applications, optional (2 wires)	Connect two leads to MA and MC. This is a relay output, which closes when the inverter output is on.

Table 4-2 – Wiring Instructions for Yaskawa V1000 Power Module

A-4 – INSTALLATION PROCEDURES

FastBack® Model 90E Technical Manual

Installation Checklist

- All conduits between platforms are flexible.
- Platforms are in accordance with equipment layout drawings.
- All conveyor bodies are level from side to side and, if possible, from front to back.
- Cross feeders are centered on dispersion tables and are at the elevation specified in equipment layout drawings.
- Distribution conveyor gates are clear of cross feeder sidewalls.
- Distribution conveyors are perpendicular to cross feeders whenever possible.
- Maximum slope is 1.5°.
- Maximum vertical gap between distribution conveyors is ¼ inch.
- Photoeye support arms are below the turned edge of the pan sidewall.
- Photoeye brackets are aimed at the apparent apex of product discharge without being obstructed by the waterfall effect of product.
- Jam nuts are tightened.
- Enclosures are sealed for watertightness. Wires are clear of seals.
- Rub points or collision points have been eliminated.